Vision-Language Geometric Programming for Long-Horizon Robotic Assembly

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***Abstract*—The abstract goes here.**

***Index Terms*—IEEE, IEEEtran, journal, LaTeX, paper, template.**

I. Introduction

THIS demo file is intended to serve as a "starter file" for IEEE journal papers produced under LaTeX using IEEEtran.cls version 1.8b and later.

II. Related Work

III. Preliminaries and Problem Statement

Traditionally, robotic task execution relied heavily on hard-coded trajectories, which severely limited system generalization. The advent of Task and Motion Planning (TAMP) significantly lowered the deployment barrier by abstracting continuous motion into symbolic predicate logic, such as the Planning Domain Definition Language (PDDL) [11]. However, in long-horizon tasks, this framework remains strictly dependent on human experts to manually and exhaustively define all preconditions and effects. This zero-tolerance for logical incompleteness makes manual specification highly susceptible to critical omissions.

To alleviate this manual bottleneck, recent advancements have integrated Vision-Language Models (VLMs) as semantic front-ends to autonomously generate TAMP domain rules [8]. While this greatly enhances system usability, such pure semantic-driven architectures suffer from inherent logical brittleness. Because Large Language Models (LLMs) operate without underlying physical grounding, they are prone to logical discontinuities during long-horizon reasoning. For instance, an LLM might generate a high-level command to "grasp the egg" while omitting the implicit physical prerequisite to "open the drawer" first. Without a geometric feedback mechanism, such semantic gaps inevitably lead to cascading execution failures.

Theoretically, Logic-Geometric Programming (LGP) [1] serves as the optimal paradigm to bridge this semantic-to-physical gap, demanding an even lower task specification threshold than VLM-guided TAMP. By tightly coupling discrete symbolic logic and continuous trajectory optimization into a single joint objective function, LGP only requires an abstract terminal goal (e.g., `(on pot egg)`). When the underlying continuous solver attempts to penetrate a closed drawer's collision mesh to reach the egg, the kinematic infeasibility naturally triggers geometric backtracking. Driven purely by rigorous mathematical optimization and spatial constraints rather than probabilistic guessing, the system autonomously deduces and inserts the required "open drawer" action sequence.

However, applying native LGP off-the-shelf to complex real-world tasks (e.g., cooking a meal involving hundreds of sequential steps) exposes two prohibitive bottlenecks:

1) Combinatorial Explosion of the Search Space: Blindly relying on geometric backtracking to derive long-horizon logic leads to a combinatorial explosion of the state space. The computational cost of evaluating deeply nested, kinematically invalid branches renders the solver intractable.

2) Singularities in Closed Kinematic Loops: Native LGP relies on a strict tree topology for its kinematic tree, defining rigid parent-child frame relationships. While many-to-one object allocations (e.g., placing multiple eggs into a single pot) maintain a valid closed logic, multi-support assemblies (e.g., placing a rigid toy-block roof across multiple supporting pillars) introduce closed kinematic loops. Forcing dynamic one-to-many topology switches in such scenarios directly violates the tree-based assumptions of the k-order Markov Optimization (KOMO) engine [2], leading to mathematical singularities and solver deadlocks.

IV. Method

Motivated by the aforementioned logical brittleness of pure semantic planners and the computational bottlenecks of native continuous solvers, we propose **VLM-LGP** (Vision-Language Model-guided Logic-Geometric Programming). It is a hierarchical planning framework that leverages a VLM to generate semantically meaningful and horizon-reducing subgoals, which in turn tightly guide an underlying LGP solver.

A. Phase 0: Scene Initialization and Reachability Pre-check

To bridge the continuous visual observation space with the discrete symbolic domain required by the LGP solver, the pipeline first utilizes Computer Vision (CV) modules to recognize object shapes and confirm their 6-DOF poses, mapping them to physical proxies in the simulation environment (e.g., an anonymous `.g` configuration). Objects are spatially sorted based on their radial Euclidean distance from the robot base to establish a preliminary reasoning sequence.

To further enhance system robustness and avoid costly downstream failures, we introduce a *Reachability Pre-check*

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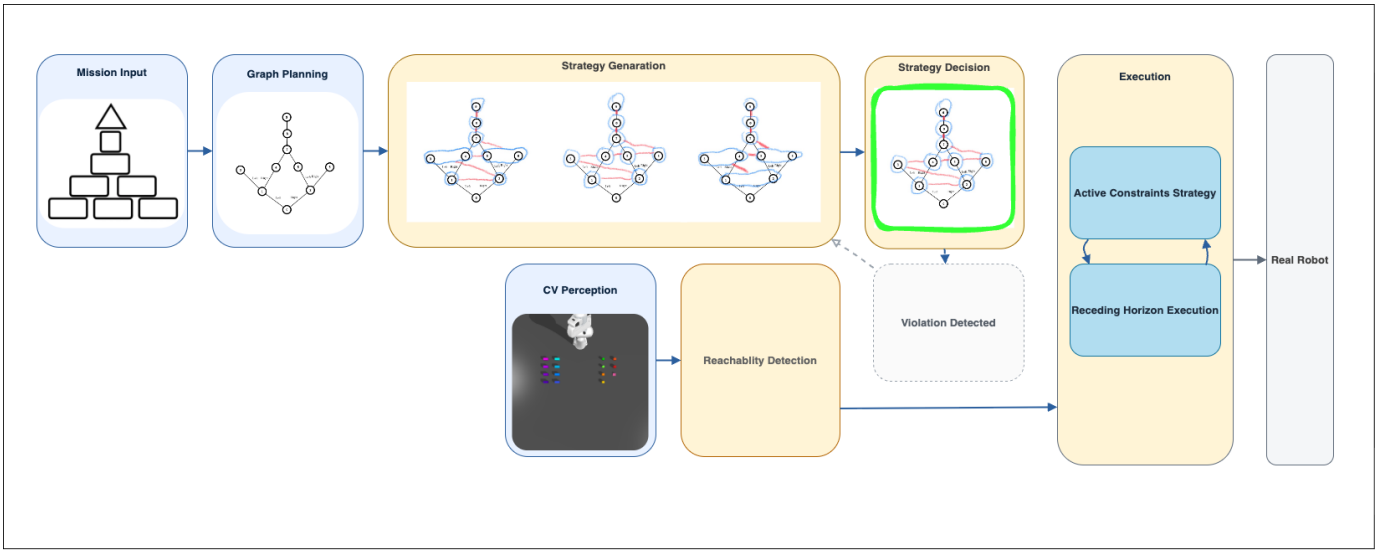

Fig. 1. The overall workflow of the proposed VLM-LGP framework.

that runs *in parallel* with the subsequent VLM reasoning phases. For each identified object o_i, the system invokes the Layer-1 LGP solver—a highly efficient waypoint generator operating at `stepsPerPhase` $= 1$—to test basic grasp feasibility under global collision constraints:

$$\text{Reach}(o_i) = \begin{cases} \texttt{true} & \text{if } \exists\, q^* : \text{KOMO}_{wp}(o_i, q^*) \text{ feasible with} \\ \texttt{false} & \text{otherwise} \end{cases} \tag{1}$$

Objects that successfully yield a valid waypoint are categorized as *reachable*; those failing or timing out are flagged as *unreachable* and excluded from the task dictionary. By leveraging this low-cost pre-filter concurrently with the semantic pipeline, the system eliminates kinematically impossible candidates before the expensive full-motion solver is ever invoked.

B. Phase 1: Extraction of the Layered Precedence Graph

In complex multi-part assembly tasks, restricting the state-space search via sequence-defining precedence graphs is an established methodology (e.g., as explored in systems like Fabrica [10]). While existing approaches often rely on dedicated physics-based disassembly planners, we propose an alternative semantic framework. Instead of requiring intensive kinematic pre-computations, a Vision-Language Model acts as a high-level zero-shot parser to process visual assembly instructions.

Similar to the sub-goal temporal decomposition seen in hierarchical planning [4], the VLM extracts a semantic *Layered Precedence Graph* $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ (see Fig. 2). Each node $v \in \mathcal{V}$ encapsulates a disjoint assembly stage (e.g., placing the foundation before the second layer), generating an ordered sequence of semantic placement tasks. This graph serves as an abstract temporal skeleton for downstream processing. The extracted graph $\mathcal{G}$ is simultaneously forwarded to a Python compliance monitor. For any directed edge $(v_i, v_j) \in \mathcal{E}$, v_i must be fully completed before any action in v_j can commence; if a downstream strategy violates this topological ordering, it is immediately rejected.

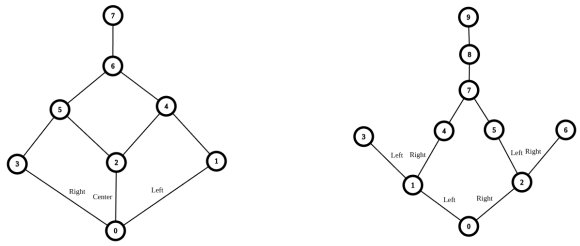

Fig. 2. Examples of the Layered Precedence Graph extracted by the VLM from visual assembly instructions (left: simple structure; right: complex structure). Edges encode strict temporal dependencies between assembly layers.

C. Phase 2: Dynamic Resource Allocation and Batch Compilation

To overcome the combinatorial explosion identified in Section III, we introduce a dynamic pruning pipeline comprising a *Semantic Resource Allocator* and a *Logic Batch Compiler*. This phase bridges the global temporal skeleton with localized execution constraints.

For each specific task node extracted from the global Layered Precedence Graph in Phase 1, the Semantic Resource Allocator first actively binds generic object requests (e.g., "cylinder") to uniquely available physical entities in the live simulation inventory (e.g., `cyl1` or `cyl2`).

Crucially, instead of blindly outputting a single static plan, the system leverages the VLM's reasoning capabilities to explicitly generate multiple candidate logic sub-graphs (execution sequences) (see Fig. 2 for illustration) that satisfy the current node's structural requirements. The VLM then acts as an autonomous evaluator, assessing these deterministic routes

against spatial heuristics—such as minimizing arm occlusion, avoiding redundant tool switches, or favoring top-to-bottom manipulation. It explicitly selects the most physically feasible trajectory and outputs the rationale behind its decision.

Before forwarding the selected strategy to the Logic Batch Compiler, the Python compliance monitor cross-validates the chosen execution sequence against the Layered Precedence Graph $\mathcal{G}$ to guarantee that no layer-ordering violations exist (see Section IV-B).

Only after this rigorous semantic filtering and logical validation is the selected optimal sequence passed to the Logic Batch Compiler. This compiler translates the chosen sub-graph into hardware-safe execution batches guided by strict dependency rules, explicitly outputting flat, atomic facts adhering to native LGP syntax (e.g., generating `(on top_left_base1 cyl1)` alongside designated action rules like `pick_cylinder`).

Consequently, by harnessing the VLM's physical understanding of the real world combined with LGP's unified continuous-discrete planning, this framework elevates the overall system robustness and success rate to a new level. Furthermore, benefiting from the rapid advancement of large language models, the success rate of the semantic reasoning phase is already exceptionally high, establishing a solid foundation for the performance comparisons presented in the experimental evaluations.

1) Waypoint-Gated Active Constraint Strategy: A notorious bottleneck in native TAMP and LGP solvers is the combinatorial explosion of collision checking when scenes contain numerous objects. Evaluating explicit collision constraints among all N environmental entities imposes an $\mathcal{O}(N^2)$ burden on the full-motion optimizer, the majority of which are geometrically irrelevant.

To resolve this, we interleave a dynamic *Active Constraint Strategy* between the solver's waypoint generation and full-motion optimization stages. Once the Layer-1 solver generates sparse waypoints for a sub-task, the system selectively activates collision pairs based on geometric proximity:

$$C_{\text{active}}(w) = \{(l_i, o_j) \mid d(l_i^w, o_j) < r_{\text{active}}\} \tag{2}$$

where l_i^w denotes the i-th robot link pose at waypoint w, and o_j is an obstacle frame. Only the object pairs within the dynamic spatial radius r_{active} are instantiated as explicit collision constraints in the LGP configuration; the vast majority of irrelevant scene objects are safely ignored. If the subsequent full-motion solve fails, the system expands r_{active} and retries, providing a principled fallback. This optimization dramatically shrinks the search manifold while strictly maintaining physical safety.

D. Phase 3: Constrained Geometric Execution

The geometric execution phase translates the logic bounds dynamically compiled in Phase 2 into continuous trajectory optimization using the native C++ LGP infrastructure [1]. Because the VLM has strictly pruned the logical permutations, the k-order Markov Optimization (KOMO) solver focuses solely on resolving local geometric feasibility. The core action skeleton is refined through three specialized, physics-grounded manipulation primitives.

1) Shape-Aware Semantic Grasping Primitives: Native LGP solvers typically rely on a single, hard-coded top-down grasping approach that targets an object's geometric centroid. This severely restricts the solver's maneuverability and frequently fails when deployed to real-robot hardware in cluttered environments. To overcome this, we design shape-specific manipulation primitives that maximize the solver's configuration-space freedom while remaining deployable on physical platforms.

For all grasp types, the system first creates a 6-DOF free virtual snap frame f_{snap} attached to the gripper, and initializes it to a semantically meaningful target pose:

$$\begin{aligned} f_{snap} &\leftarrow \text{addFrameDof}(gripper, \texttt{JT\_free}); \\ \text{pose}(f_{snap}) &\leftarrow \text{pose}(targetF) \end{aligned} \tag{3}$$

where $targetF$ is determined by traversing the object's kinematic tree (e.g., a designated handle sub-frame for large objects).

Box-type objects. The grasp constrains the gripper to be centered on the target frame in the XY plane while leaving the Z axis free for the solver to optimize:

$$\phi^x_{\text{posRel}}(gripper, targetF) = 0, \quad \phi^y_{\text{posRel}}(gripper, targetF) = 0 \tag{4}$$

A pre-grasp funnel guides the gripper from a standoff distance d_{pre} above the target before final descent, with soft orientation alignment enforced throughout the approach corridor.

Cylindrical objects. The primitive enforces strict orthogonality between the gripper's approach axis and the cylinder's central axis, while permitting translational sliding along the cylinder length:

$$\phi^{xy}_{\text{posRel}}(gripper, O) = 0, \quad \langle \hat{x}_{gripper}, \hat{z}_O \rangle = 0 \tag{5}$$

$$-\tfrac{L}{2} + m \le \phi^z_{\text{posRel}}(gripper, O) \le \tfrac{L}{2} - m \tag{6}$$

where L is the cylinder length and m a safety margin. This formulation grants the solver maximal freedom to determine the optimal grasp height along the object's axis, which is critical for collision-free manipulation in dense scenes.

2) Topological Decoupling for Multi-Support (`action_place_multi`): Native LGP assumes strict open-chain kinematic trees, which fail in multi-support structural scenarios. We resolve this by introducing a highly adaptable unactuated 6-DOF *Virtual Anchor* (A_v) parented directly to the world frame.

For placing an object O onto a composite set of support frames $S_{1..n}$, A_v is assigned an initial pose matching the calculated target centroid. The solver first locks the anchor to the computed target world pose via a hard equality constraint, then employs a soft alignment to guide the object towards the anchor:

$$\Phi_{\text{lock}} = \|\text{pose}(A_v) - p_{\text{target}}\|^2_{eq} = 0 \tag{7}$$

$$\Phi_{\text{align}} = \|\phi_{\text{poseDiff}}(A_v, O, x_t)\|^2_{sos} \to \min \tag{8}$$

A rigid kinematic switch then transfers ownership of O to A_v, elegantly breaking the closed kinematic loop and granting the

KOMO solver dynamic capability to iteratively interpolate the most stable geometric descent.

3) Trajectory Shaping and Phased Collision Protocol: To overcome solver deadlocks caused by static long-horizon obstacles, all generated sequences are regularized through dynamic trajectory shaping:

Pre-Action Funnels: To prevent lateral sliding collisions during delicate placements, we inject relative positional waypoints at intermediate time steps t_{far} and t_{near} before the action time t. This constructs a virtual "funnel" forcing the payload precisely above the target with decreasing vertical standoff distances d_z^{far} and d_z^{near}:

$$\Phi_{\text{funnel}} = \|\phi_{\text{posRel}}(O, A_v, x_{t_{near}}) - \begin{bmatrix}0 & 0 & d_z^{near}\end{bmatrix}^T\|_{sos}^2 \quad (9)$$

The specific time offsets and standoff values are shape- and action-dependent.

Phased Collision Relaxations: Continuous collision constraints $FS_{negDistance}$ between moving gripper parts and static obstacles are segmented temporally with parameterized safety margins d_{safe}. During the *transit phase* ($[t_{start}, t_{transit}]$), strict safety margins are enforced via per-pair distance checks. Within the *final approach phase* ($[t_{transit}, t]$), these margins are hierarchically relaxed to allow physical manifold contact, enabling deadlock-free grasping and precision placement. The exact time boundaries and margin values vary across manipulation primitives to accommodate different geometric and kinematic requirements.

E. Phase 4: State Verification and Safety Halting

Given the inherent uncertainties of physical execution (e.g., slipping, grasping failures), open-loop trajectory execution is risky for long-horizon assembly. To address this, we introduce a *Visual-Semantic Validator* that performs state verification after each executed batch.

The validator captures a top-down visual observation of the current workspace and utilizes computer vision, combined with the VLM, to detect geometric deviations against the expected blueprint state established in Phase 1. If an error is detected—such as a missing component in a designated slot or an unexpectedly placed object (an "intruder")—the system immediately halts the standard planning pipeline. While dynamic closed-loop replanning and autonomous physical recovery remain theoretical extensions for future work, this rigorous error detection mechanism effectively prevents minor execution anomalies from cascading into catastrophic structural failures, establishing a critical safety boundary for the system.

(Reviewer note: This verification and halting module ensures execution integrity during long-horizon assembly. Depending on the final review focus, this section may be expanded with quantitative recovery metrics or condensed to focus on the planning-level optimizations.)

V. Experimental Results

A. Performance Evaluation

Experimental data and baseline comparisons with native LGP go here.

VI. Conclusion

The conclusion summarizes the contributions of VLM-LGP in long-horizon assembly tasks.

Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Acknowledgment

The authors would like to thank...

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